Amrita Vishwa Vidyapeetham
Amrita School of Computing, Amritapuri
BTech Computer Science and Engineering (AI)

22AIE311 - Software Engineering (Project-based)

LAB SHEET 1

CO	Course Outcomes
CO01	Understand the basic principles of Software Engineering.
CO02	Understand how to choose the appropriate SDLC models depending on the user requirements.
CO03	Understand the concept of Requirements Engineering.
CO04	Understand the concept of Software Design.
CO05	Understand how to apply the knowledge, techniques, and skills in the development of software and its maintenance.

Requirement Engineering – Inception and Elicitation Phase

Objective

To understand and perform the Inception and Requirement Elicitation activities of Requirement Engineering for a given software project.

Group Number: 4

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PART A – INCEPTION PHASE

Task 1: Problem Identification [CO03]

1. Define the problem clearly.

The rapid digitisation of academic and government records has increased the risk of document forgery, tampering and AI-generated manipulation. Existing document verification systems are centralised and manual. This makes them vulnerable to fraud and incapable of detecting subtle alterations or synthetic content effectively.

Currently, there are no secure, tamper-proof and transparent systems that combine blockchain with AI-based fraud detection to ensure document authenticity.

Therefore, a secure system is required to authenticate digital documents, detect fraud, and maintain an immutable audit trail for verification purposes.

2. What existing issues does the system solve?

The system solves the following issues:

- Fake certificates and forged government documents
  Fake academic degrees, identity documents, and government-issued certificates are frequently fabricated using editing tools. They are often difficult to detect through manual inspection alone.
- Tampered PDFs, like edited academic transcripts
  Marks, grades, issue dates, and signatures can be altered digitally without leaving obvious visible traces. Traditional verification systems lack automated mechanisms to detect such subtle modifications.
- AI-generated or synthetically altered content
  Text, signatures, and document layouts can be artificially created or manipulated. Existing systems may not be well-equipped to identify AI-generated or synthetic document content.
- Manual and time-consuming verification processes
  Institutions and employers often rely on email confirmations or physical record checks, which are slow, inefficient, and prone to human error.
- Absence of an immutable audit trail for document history
  Most centralised systems do not maintain tamper-proof logs. Document history, revocations, or modifications can be altered without detection, reducing accountability.
- Lack of transparency in centralised record-keeping systems.
  Centralised databases are vulnerable to insider manipulation and single-point failures. There is limited visibility into how documents are issued, verified, or modified.

Task 2: Identify Stakeholders

Sl. No	Stakeholder Name	Role	Interest in System
1	Students/Citizens	Document Owners	Wants easy verification of documents
2	Academic Institutions / Government/ Companies	Document Issuers	Issue authenticated documents.
3	System Administrator	System Manager	Maintain system integrity and access control
4	Blockchain Admin	Network Controller	Maintain the blockchain network
5	Security Analyst	Security Monitoring	Monitor suspicious activity on the systems or network.
6	Developers	System Builders	Design, implement and maintain the system

Task 3: Define System Scope

1. What the system will do

- *Document fingerprinting* : Upload government or academic documents through a web portal from which the metadata is extracted and a unique cryptographic fingerprint (hash) is generated for each document. This hash is stored on a permissioned blockchain, ensuring that once a document is registered, it cannot be modified or tampered with.
- *Fraud and tampering detection* : Machine learning and digital forensic methods to detect tampering of a document. It specifically detects AI-generated text or synthetic content to ensure document authenticity. The system also monitors for metadata inconsistencies and abnormal behavior patterns, such as repeated submissions
- *Verification and Validation* : Trusted authorities can issue or validate the documents. Verifiers include employers can upload a document or read a unique code to compare its current hash against the record stored on the blockchain.
- *Storage* : Encrypted document files are stored off-chain in secure object storage to maintain efficiency, while only the hash proofs are kept on the blockchain.
- *Audit* : Every actions from issuance to verification is logged in a tamper-proof audit trail for full transparency and traceability

2. What the system will NOT do

- *No Hardware Interfacing* : It is not designed to interface with physical scanners or hardware-based forensic tools (e.g., UV light scanners) to detect paper-based forgeries.
- *Pre-Blockchain History* : The system cannot verify the history of a document *'before'* it was first registered on the platform. It can only guarantee immutability from the moment the hash is recorded on the blockchain.

- *Authenticating False Data from Trusted Sources* : If a legitimate, authorized "Issuer" (like a university) uploads a document containing factually incorrect information (e.g., a typo in a grade), the system will verify the document as *authentic* because it came from a trusted source. It verifies *source and integrity*, not necessarily the *factual truth* of the content provided by the issuer.
- *Legal Decisions* : While the system provides an "Authenticity Score", it will not replace the final legal judgment of a forensic expert or a court of law. It acts as an *assistive tool* to highlight risks, but final high-stakes decisions still require human oversight.
- *No Subjective Interpretation* : The AI will not interpret the *meaning* of a document (e.g., whether a person is *qualified* for a job); it strictly checks for *technical tampering* and *synthetic generation*.
- *Offline Verification Without QR* : The system cannot verify a document's authenticity without a network connection to the blockchain, unless a specialized QR code cached (like in a Flutter app) is used.
- *Auto Entry of Old Records* : The system will not automatically pull data from a database and verify. It requires the institution to participate and 'register' the documents manually.

3. System boundaries

- *Authorized Issuers* : The system only accepts "authentic" registrations from pre-verified institutions (like Amrita University or government agencies). It is not an open ledger for any user to "certify" their own documents without a validator in the loop.
- *Digitized Format Only* **:** The system boundary begins at the point of digital ingestion. It only processes electronic files (PDFs and high-resolution images). It does not extend to the physical handling of paper documents or the hardware used to scan them.
- *Identity Scope* **:** The system manages document-related metadata and cryptographic proofs. It is not a full-scale Human Resource Management System (HRMS) or a Student Information System (SIS); it merely interfaces with them for verification.
- *Internet Dependency* : The system operates within the boundary of an active network connection. While a Flutter app might offer offline QR verification, the "Source of Truth" (the blockchain) remains outside that offline environment.
- *Assistance vs. Authority* **:** The AI & Forensics module acts as a "decision support system." Its boundary ends at providing an *Authenticity Score*. The legal authority to act on that score (e.g., rejecting a job applicant) lies with the human stakeholder.
- *Data Retention* **:** The system is bound by *GDPR/Data Privacy laws*. While the blockchain record is permanent, the off-chain document storage is subject to "Right to Erasure" policies, meaning the file can be deleted even if the cryptographic hash remains on the ledger.

Task 4: Identify Constraints

1. Budget constraints

Since this is an academic project, the system can only operate with limited finance resources. This restricts the use of enterprise-grade blockchain infrastructure, paid AI APIs or forensic detection tools and commercial cyber security services.

2. Time constraints

The project needs to be designed, developed, tested and documented within a single academic semester. This restricts the time available for extensive requirement validation, limits ability to train custom AI models on large datasets and also reduces the scope for large-scale user testing and performance benchmarking.

3. Technical constraints

This project integrates advanced technologies, which introduces several challenges. Public blockchain platforms have high transactional costs and network congestion may affect real-time document registration. AI-based fraud detection models may produce more false positives and false negatives if not trained well and optimised. Furthermore, due to privacy restrictions, access to real academic or government document datasets is limited, leading to reliance on publicly available datasets that may not accurately represent real-world fraud patterns.

4. Legal constraints

Since the system handles sensitive documents, it must adhere to the legal requirements. PErsonaldata should be protected in accordance with the data protection laws. Academic certificates, identification documents, and official government records contain confidential information, and any improper storage or unauthorised access may result in legal liability. Additionally, the system must comply with IT and cybersecurity regulations, to ensure that blockchain-based records are legally recognised as valid proof of document integrity.

Task 5: Feasibility Study

1. Technical Feasibility:

The project is technically feasible using existing technologies and machine learning libraries for AI-based fraud detection. Other open-source tools like OpenCV, PyMuPDF, etc. can support document analysis. Permissioned blockchain deployment and modular microservices architecture make it easier to implement.

2. Economic Feasibility

The project relies mostly on open-source technologies and minimal cloud deployment, therefore the development cost is low. Hosting can be performed using affordable cloud platforms or local servers. Therefore, the project is economically feasible for academic implementation.

3. Operational Feasibility

Provides an intuitive web interface with role-based dashboards for users, issues, and verifiers. Automation reduces manual verification workload and improves efficiency. Different institutions can adopt the system with minimal training. This proves that the system is operationally feasible.

4. Schedule Feasibility

The modular architecture of the system allows a phased implementation. Frontend, backend, AI module, and blockchain integration can be developed incrementally. With proper task division among team members, the project can be streamlined and completed in the required timeline.

PART B – REQUIREMENT ELICITATION PHASE

Task 6: Select Elicitation Techniques [CO03]

*Choose at least **two techniques**:*
1. Interview
2. Questionnaire
3. Brainstorming
4. Observation
5. Prototyping

Explain why you selected them.

From the list of given techniques, we have chosen Interview and Brainstorming as our elicitation techniques.

- *Interview:* This technique is used to gather detailed information directly from the stakeholders. Through structured or semi-structured interviews, the development team can understand the challenges that are frequently faced in the document verification process, the limitations of existing systems, and the expectations that are to be satisfied when building a secure verification platform. This technique is helpful in identifying real-world requirements like why we need faster verification, fraud detection capabilities, and secure record storage. They also allow the stakeholders to explain their workflows and pain points. Overall, it helps the development team design features that address practical problems.
- *Brainstorming:* It is used within the development team and with domain experts to generate ideas and potential solutions for the system. In this process, team members discuss different approaches for blockchain integration, fraud detection, document fingerprinting, and verification workflows. This technique promotes collaborative thinking and helps identify innovative features, refine system architecture, and prioritise the most important functionalities required for the project,

Task 7: Prepare Interview Questions (Minimum 8 Questions)

1. What are the challenges you currently face in verifying the documents?
2. What security features do you expect in a document verification system?
3. Who are the primary users of the system?
4. What type of interface would be most convenient for the users(mobile/web/or both)?
5. How useful is QR-code based verification for fast document verification?
6. What types of documents do you frequently need to verify?
7. Are there any digital systems for document verification and if so, how does it work?
8. What type of analytics would be useful for monitoring the document verification process?
9. What level of access control should different users have in the system?
10. Who can have the authority to revoke or update a document record?
11. What main features should the system have?
12. What response time would be acceptable for the document verification results?
13. What type of information should be stored along with the document?
14. What level of privacy protection is required for sensitive documents?
15. Should the system integrate with existing institutional or government databases?

Task 8: Identify Requirements

1. Functional Requirements

- *Document Registration & Hashing*: The system must allow authorized issuers to upload documents, extract metadata, and generate unique cryptographic hashes.
- *Blockchain Integration*: The system must commit document hashes to a permissioned blockchain (Ethereum/Polygon) to create an immutable record of issuance.
- AI-Driven Tamper Detection: The forensics module must analyze uploaded files for image manipulation (pixel inconsistencies) and metadata anomalies. Preferably use openCV and ExifTool
- *Synthetic Content Identification*: The system must utilize Transformer-based models to calculate perplexity and burstiness to identify AI-generated text.
- *Verification Service*: The system must provide a public-facing portal or QR code interface where verifiers can compare a document's current hash against the blockchain record.
- *Role-Based Access Control (RBAC)*: The system must differentiate permissions between Citizens (uploaders), Issuers (authorizers), Verifiers, and Admins.
- *Fraud Analytics*: The system must track submission patterns (using Isolation Forests) to flag suspicious behavioral clusters or repeated forgery attempts.
- *Notification & Reporting:* The system must generate a detailed "Authenticity Score" and risk report for every verification request.

2. Non-Functional Requirements (System quality attributes)

- *Immutability*: Once a document hash is recorded on the blockchain, it must be impossible for any user, including administrators, to alter or delete the record.
- *Security*: All document transfers must be encrypted using AES/SSL protocols, and off-chain storage (MinIO/S3) must be isolated from public access.
- *Scalability*: The microservices architecture (FastAPI) and hybrid storage model must support an increasing volume of document registrations without degrading performance.
- *Availability*: The system must ensure that the verification portal remains accessible even if individual microservices or blockchain nodes experience downtime.
- *Privacy*: The system must adhere to data protection standards by storing only cryptographic proofs on-chain, ensuring that Personally Identifiable Information (PII) can be deleted from off-chain storage if requested.
- *Interoperability*: The API Gateway must allow seamless integration with existing institutional databases and third-party verification tools.
- *Auditability*: Every transaction—from issuance to verification—must leave a transparent, time-stamped trail on the ledger for forensic auditing.

Task 9: Prepare Use Case List

Use Case ID	Use Case Name	Actor
UC-01	Register/Login to the system	User / Issuer / Verifier / Admin
UC-02	Upload Document	User / Issuer
UC-03	Enter Document Metadata	User / Issuer
UC-04	Generate Document Fingerprint	System
UC-05	Perform AI and Forensic Analysis	System
UC-06	View authenticity score and risk report	User / Issuer / Verifier
UC-07	Register Document on Blockchain	Issuer
UC-08	Generate QR Verification code	System
UC-09	Verify the document during upload	Verifier
UC-10	Verify the document using the QR code	Verifier
UC-11	Compare Document Hash with Blockchain records	System
UC-12	Check Issuer Authenticity and Revocation Status	System / Verifier
UC-13	Store document is secure off-chain storage	System
UC-14	View verification result	User
UC-15	Maintain Audit Log	Admin
UC-16	Monitor Suspicious Activities	Admin
UC-17	Revoke Document	Issuer
UC-18	Manage Users and Roles	Admin